(20)

Ex:-3- $A=\frac{1}{20}$, $B=\frac{1}{30}$, $C=\frac{1}{40}$ ⇒ together ga how many days?

Sol:- $\frac{1}{20}+\frac{1}{30}+\frac{1}{40} = A+B+C = \frac{6+4+3}{120} = \frac{13}{120}$ = Combeffs

together days $= \frac{1}{\text{Combeffs}} = \frac{120}{13} = 9\frac{3}{13}$ days.

(or)

T.N.D $= \frac{LCM(20,30,40)}{\frac{LCM}{20}+\frac{LCM}{30}+\frac{LCM}{40}} = \frac{120}{\frac{120}{20}+\frac{120}{30}+\frac{120}{40}} = 9\frac{3}{13}$ days

(or) ⇒ Same quickly

20, 30, 40

$\frac{120}{6+4+3} = 9\frac{3}{13}$ days

(or) Time taking

formula method:- when 3 members are given

together $= \frac{xyz}{xy+yz+zx} = \frac{20\times30\times40}{20\times30+30\times40+40\times20}$

$= \frac{24000}{600+1200+800} = \frac{240}{26} = \frac{120}{13} = 9\frac{3}{13}$ days

Note:- Wages will be given according to portion of work completed by individual persons.

Wages are devided according to their efficiencies with respect to their completed days.

Ex:-4:- 6000 to be distributed to workers A & B. If $A=\frac{1}{20}$, $B=\frac{1}{30}$, their wages are?

T.N.D $= \frac{xy}{x+y} = \frac{20\times30}{20+30} = 12$ days.

A eff for 12 days $= 12\times\frac{1}{20}$ = Part of work by A for 12 days

B eff " " $= 12\times\frac{1}{30}$ = " " B " "

(61)

Wage of A = Part of work × Total Amount

$= \frac{12}{20} \times 6000 = 12 \times 300 = 3600/-$.

Wage of B = Part of work by B × Total amount

$= \frac{12}{30} \times 6000 = 12 \times 200 = 2400/-$.

↳ Part of work:- Work done by a person in some days of total is Part of work.

Ex:- A can complete a work in 10 days. then

Part of work in 5 days =. 5 × one day work

$= 5 \times$ efficiency $= 5 \times \frac{1}{10}$

$= \frac{1}{2}$.

Part of work in n days =. n × one day work

Part of work in 'n' days = no. of days × efficiency

↳ Usage of Part of work:- In general in real-time scenarios total work together not seen. But we used to see few people are joining & few people are leaving. In these cases part of work is used.

Ex:- A, B & C started the work. After some days B left the job and after some more days A left the job. the remaining work completed by C. Then total no. of days ?

→ In the above case all 3 completed the work (let us '1')

but not together means part of work by A, by B & by C.

∴ Part of A + Part of B + Part of C = 1 (Full work)

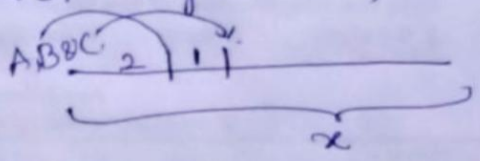
$mA + nB + oC + \dots = 1$

Where $m, n, o \neq 0$ are no.of days worked by A, B & C.

" " A, B & C " efficiencies of A, B & C.

In general total work is to be consider as 'x' and all other ($m, n, o \neq 0$) are considered in terms of 'x'. to find 'x'.

Ex-5:- $A = \frac{1}{12}$, $B = \frac{1}{24}$, $C = \frac{1}{36}$. A, B & C started the work. After 2 days B left and C left after 1 day of B left. Total no.of days work done is ?



Ans:- Let's assume total days as 'x'

→ no.of days done by 'A' is 'x' because he is working till the end since beginning.

→ no.of days done by 'B' is '2'. After 2 days working B left.

→ no.of days done by 'C' is '3', After 1 day of B left means B working days + 1 days = 2+1 = 3 days.

$$mA + nB + oC = 1$$

$$x \times \frac{1}{12} + 2 \times \frac{1}{24} + 3 \times \frac{1}{36} = 1$$

$$\Rightarrow \frac{x}{12} + \frac{1}{12} + \frac{1}{12} = 1 \Rightarrow x + 2 = 12 \Rightarrow \boxed{x = 10 \text{ days}}$$

↳ Chain Rule (or) Equalizing efficiencies:-

→ here in this we have to consider every individual effort is same irrespective of how much work that individual is doing and is to be given same wage. [workers in construction of buildings].

→ Suppose consider construction of two buildings. Means workers will be given each same in both cases.